PROBLEM CL4

Cascade of Aerofoils: Downstream State

Original statement · CL4.pdf

Periodicity | Mass, momentum, energy | Perfect gas | Cascade

OFFICIAL PROBLEM STATEMENT

A planar uniform stream of air with pressure p_1, density ρ_1 and velocity $\bar{v}_1 = v_{1x}\bar{e}_x$ flows past an array of aerofoils with vertical separation L, exerting on each one a force (per unit length) given by $\bar{F} = F_x\bar{e}_x + F_y\bar{e}_y$. Assuming that the aerofoils are thermally insulated and neglecting the effect of gravity, write the equations that determine the values of the pressure p_2, density ρ_2 and velocity $\bar{v}_2 = v_{2x}\bar{e}_x + v_{2y}\bar{e}_y$ for the uniform stream found sufficiently far downstream.

Source: CL4.pdf (page 1). Notes.pdf, Chapter 3: integral mass balance Eq. 3.11, integral momentum balance Eq. 3.36, integral total energy balance Eq. 3.46, force on a body Eq. 3.32.

Worked solution (4-phase method)

PHASE 1

Hypotheses, Data and Unknowns

- Steady planar flow through an infinite array of identical aerofoils (pitch L), so the flow is periodic in y; uniform far upstream (1) and far downstream (2).
- Neglect $\bar{g}$; aerofoils thermally insulated; compressible fluid. Viscous stresses and heat conduction negligible in the uniform sections.
- Perfect gas: $h = e + \frac{p}{\rho} = c_pT = \frac{c_p}{R_g}\frac{p}{\rho} = \frac{\gamma}{\gamma-1}\frac{p}{\rho}$ (with $p = \rho R_g T$ and $\gamma = c_p/c_v$), as in the class solution.
- Data: $[p_1,\ \rho_1,\ v_{1x},\ L,\ F_x,\ F_y,\ \gamma]$. Unknowns: $[p_2,\ \rho_2,\ v_{2x},\ v_{2y}]$: four equations are needed.

Symbol	Meaning	SI unit
p_1, ρ_1, v_{1x}	upstream state	Pa, kg/m^3, m/s
$p_2, \rho_2, v_{2x}, v_{2y}$	downstream state	Pa, kg/m^3, m/s

Symbol	Meaning	SI unit
L	pitch of the array	m
F_x, F_y	force on each aerofoil per unit length	N/m

PHASE 2

Frames, Control Volume and Surfaces

- Frame fixed to the array; $\bar{e}_x$ along the incoming stream, $\bar{e}_y$ along the array. Fixed control volume, $\bar{v}_c = 0$, one period high.
- Surfaces: Σ_1 (far upstream, height L, $\bar{n} = -\bar{e}_x$), Σ_2 (far downstream, height L, $\bar{n} = +\bar{e}_x$), Σ_{l1} and Σ_{l2} (upper and lower stream surfaces, $\bar{v} \cdot \bar{n} = 0$), Σ_w (aerofoil surface, $\bar{v} = 0$, insulated).
- Periodicity: Σ_{l2} is Σ_{l1} shifted by $-L\bar{e}_y$ with opposite normals, so the integrals of any periodic quantity over Σ_{l1} and Σ_{l2} cancel each other out.

Force on each aerofoil (per unit length), with $\bar{n}$ the outward normal of the control volume:

$$\boxed{\bar{F} = F_x \bar{e}_x + F_y \bar{e}_y = \int_{\Sigma_w} p\,\bar{n}\,d\sigma - \int_{\Sigma_w} \bar{\bar{\tau}}' \cdot \bar{n}\,d\sigma}$$

PHASE 3

Balances, Surface by Surface

Mass (Notes Eq. 3.11)

$$\underbrace{\frac{d}{dt}\cancel{\int_{V_c} \rho\,dV}}_{0\ \text{(steady)}} + \int_{\Sigma_c} \rho\,(\bar{v} - \underbrace{\cancel{\bar{v}_c}}_{0})\cdot \bar{n}\,d\sigma = 0$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho\,\bar{v}\cdot\bar{n}\,d\sigma$
Σ_1	$\rho_1(-v_{1x})\,L$
Σ_2	$\rho_2\,v_{2x}\,L$
$\Sigma_{l1},\ \Sigma_{l2}$	0 (stream surfaces)
Σ_w	0 (wall)

$$\boxed{\rho_1 v_{1x} = \rho_2 v_{2x}} \qquad (1)$$

Linear momentum (Notes Eq. 3.36)

$$\underbrace{\frac{d}{dt}\int_{V_c}\rho\bar{v}\,dV}_{0\ \text{(steady)}} + \int_{\Sigma_c}\rho\bar{v}\,[(\bar{v}-\underbrace{\bar{v}_c}_{0})\cdot\bar{n}]\,d\sigma = -\int_{\Sigma_c} p\,\bar{n}\,d\sigma + \int_{\Sigma_c}\bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma + \underbrace{\int_{V_c}\rho\bar{g}\,dV}_{\text{neglect }\bar{g}}$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho\bar{v}\,(\bar{v}\cdot\bar{n})\,d\sigma$	$-\int p\,\bar{n}\,d\sigma$	$\int \bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma$
Σ_1	$\rho_1 v_{1x}\bar{e}_x(-v_{1x})L$	$+p_1 L\,\bar{e}_x$	0
Σ_2	$\rho_2(v_{2x}\bar{e}_x + v_{2y}\bar{e}_y)\,v_{2x}L$	$-p_2 L\,\bar{e}_x$	0
$\Sigma_{l1}+\Sigma_{l2}$	0	0 (cancel each other out)	0 (cancel each other out)
Σ_w	0	$-\int_{\Sigma_w} p\,\bar{n}\,d\sigma$	$\int_{\Sigma_w}\bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma$

Boxed Σ_w terms: $-\bar{F} = -F_x\bar{e}_x - F_y\bar{e}_y$. With (1), $\rho_2 v_{2x}L = \rho_1 v_{1x}L$ in the Σ_2 flux.

$$\boxed{F_x = \rho_1 v_{1x} L\,(v_{1x}-v_{2x}) + (p_1-p_2)\,L} \quad (2) \qquad \boxed{F_y = -\rho_1 v_{1x} L\,v_{2y}} \quad (3)$$

Total energy (Notes Eq. 3.46)

$$\underbrace{\frac{d}{dt}\int_{V_c}\rho\left(e+\frac{v^2}{2}\right)dV}_{0}+\int_{\Sigma_c}\rho\left(e+\frac{v^2}{2}\right)(\bar{v}\cdot\bar{n})\,d\sigma=-\int_{\Sigma_c}p\,\bar{v}\cdot\bar{n}\,d\sigma+\int_{\Sigma_c}\bar{v}\cdot(\bar{\bar{\tau}}'\cdot\bar{n})\,d\sigma+\underbrace{\int_{V_c}\rho\bar{g}\cdot\bar{v}\,dV}_{\text{neglect}}-\int_{\Sigma_c}\bar{q}\cdot\bar{n}\,d\sigma+\underbrace{\int_{V_c}(Q_c+Q_r)\,dV}_{0}$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int\rho\left(e+\frac{v^2}{2}\right)(\bar{v}\cdot\bar{n})\,d\sigma$	$-\int p\,\bar{v}\cdot\bar{n}\,d\sigma$	$\int\bar{v}\cdot(\bar{\bar{\tau}}'\cdot\bar{n})\,d\sigma$	$-\int\bar{q}\cdot\bar{n}\,d\sigma$
Σ_1	$\rho_1\left(e_1+\frac{v_{1x}^2}{2}\right)(-v_{1x})L$	$+p_1v_{1x}L$	0	0
Σ_2	$\rho_2\left(e_2+\frac{v_{2x}^2+v_{2y}^2}{2}\right)v_{2x}L$	$-p_2v_{2x}L$	0	0
$\Sigma_{l1}+\Sigma_{l2}$	0	0	0 (cancel each other out)	0 (cancel each other out)
Σ_w	0	0 ($\bar{v}=0$)	0 ($\bar{v}=0$)	0 (insulated)

Moving the pressure work to the left turns e into $h=e+p/\rho$; dividing by $\rho_1v_{1x}L=\rho_2v_{2x}L$ and using $h=\frac{\gamma}{\gamma-1}\frac{p}{\rho}$:

$$\boxed{\frac{\gamma}{\gamma-1}\left(\frac{p_2}{\rho_2}-\frac{p_1}{\rho_1}\right)+\frac{v_{2x}^2+v_{2y}^2}{2}-\frac{v_{1x}^2}{2}=0}\qquad(4)$$

(1) to (4) are four equations for the four unknowns p_2, ρ_2, v_{2x}, v_{2y}, which answers the question.

PHASE 4

Interpretation, Limits and Dimensional Check

- (3): the transverse force on an aerofoil equals minus the y momentum given to the stream; the flow is deflected opposite to the force. (2) splits the streamwise force into a pressure drop and a momentum change.
- $\bar{F}=0$: (1) to (4) are satisfied by state 2 = state 1. Incompressible limit ($\rho_2=\rho_1$): $v_{2x}=v_{1x}$ and $p_1-p_2=F_x/L$.
- Dimensions (SI): $[F_x/L]=\mathrm{Pa}=[\rho_1v_{1x}^2]$; $[\rho_1v_{1x}Lv_{2y}]=\mathrm{N/m}$; $[\gamma p/((\gamma-1)\rho)]=\mathrm{m^2/s^2}=[v^2]$.
- Agreement with the official solution and the class solution: the same four equations (1) to (4).

Extension (beyond the official solution): closed solution

With $m = \rho_1 v_{1x}$: $\rho_2 = m/v_{2x}$, $v_{2y} = -F_y/(mL)$ and $p_2 = P - m\,v_{2x}$ with $P = p_1 + mv_{1x} - F_x/L$. Substituting into (4), with $E = \frac{\gamma}{\gamma-1}\frac{p_1}{\rho_1} + \frac{v_{1x}^2}{2}$ and the coefficient $\frac{1}{2} - \frac{\gamma}{\gamma-1} = -\frac{\gamma+1}{2(\gamma-1)}$ of v_{2x}^2: $(\gamma+1)v_{2x}^2 - 2\gamma\frac{P}{m}v_{2x} + 2(\gamma-1)\left(E - \frac{v_{2y}^2}{2}\right) = 0$. For $\bar{F} = 0$ one root is $v_{2x} = v_{1x}$; the root continuous with it is the physical one. Check: $p_1 = 101325$ Pa, $\rho_1 = 1.2$ kg/m^3, $v_{1x} = 100$ m/s, $L = 0.2$ m, $F_x = 30$ N/m, $F_y = -80$ N/m, $\gamma = 1.4$ give $v_{2x} = 100.16$ m/s, $v_{2y} = 3.3333$ m/s, $\rho_2 = 1.19808$ kg/m^3, $p_2 = 101155.8$ Pa, residuals of (1) to (4) below 10^{-9}; the other root gives $p_2 = -6843.3$ Pa < 0 and is discarded.